

Assessing the Validity of Pressurized Vessel Impulse Approximations

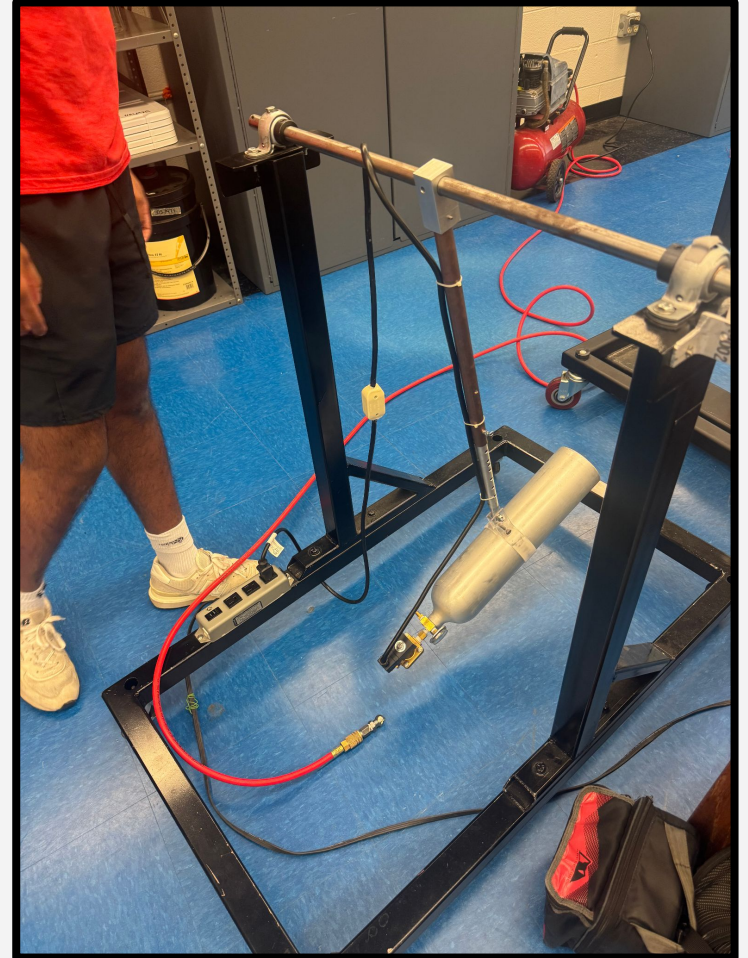
MAE 3840: Mechanical Engineering Lab

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Introduction

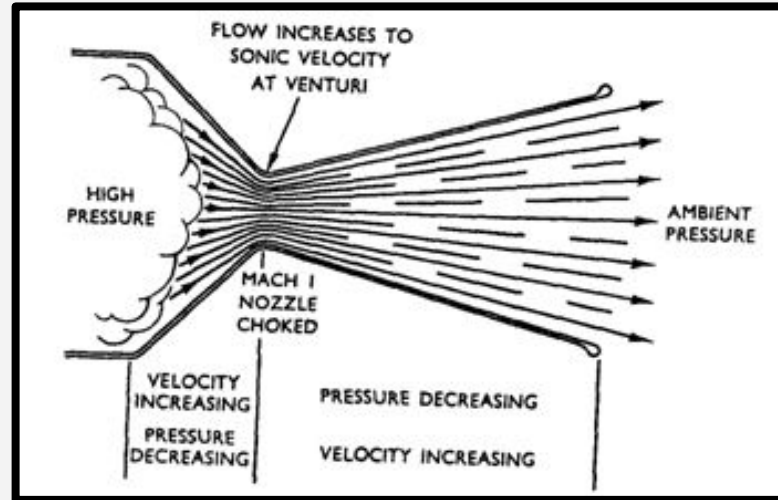
The goal of this experiment is to assess the validity of a model that **predicts** the impulse generated from the discharge of a **pressurized vessel**.

This experiment will focus on testing whether a **choked flow assumption** is an appropriate assumption and approach to take in the creation of a predictive model.



Hypothesis

The team predicts that developing a model under the assumption of choked flow **will produce accurate results** because, intuitively, most of the force supplied to the system would occur during the initial discharge while the air is escaping at its highest velocity and mass flow rate.



Background Theory

Our model is based on the general Thrust Equation, approximating the force over time based on mass flow rate, velocity, pressure, and area at the nozzle, releasing to an external pressure.

$$F(t) = \frac{dm}{dt} v_*(t) + [P_*(t) - P_g] A_*$$

By using choked flow, we can assume velocity is Mach 1 for our fluid, dry air, ~343 m/s. Area of the nozzle can be measured physically, and our choked flow assumption can also be derived to provide models for mass flow rate and the pressure at the nozzle.

$$\frac{dm}{dt} = C_d A_* P_0 \sqrt{\frac{\gamma}{R_s T_0} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}}$$

C_d is discharge coefficient, selected based on the geometry of the nozzle. P_0 is initial tank pressure, T_0 initial temperature, γ is specific heat ratio for dry air, ~1.4, and R_s is the specific gas constant of dry air, ~287.05 J/(kg*K).

Two Routes: Isothermal & Adiabatic

When determining the tank pressure and temperature's curves, two approaches can be taken. Under an adiabatic assumption, as the pressure decreases internally, it's assumed no heat transfer occurs between the air and the walls and energy is conserved. Under an isothermal assumption, the release of pressure is believed to occur slowly enough that the internal temperature can remain constant. This results in the following two models for impulse:

Adiabatic Impulse:

$$J = \int_0^t \left(A_* \left[P_0 \left(1 + \frac{\gamma-1}{2} \frac{t}{\tau} \right)^{\frac{2\gamma}{\gamma-1}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}} (\gamma C_d + 1) - P_g \right] \right) dt$$

Isothermal Impulse:

$$J = \int_0^t \left(A_* \left[P_0 e^{-\frac{t}{\tau}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}} (\gamma C_d + 1) - P_g \right] \right) dt$$

Where τ is:

$$\tau = \frac{V}{C_d A_* c_0} \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

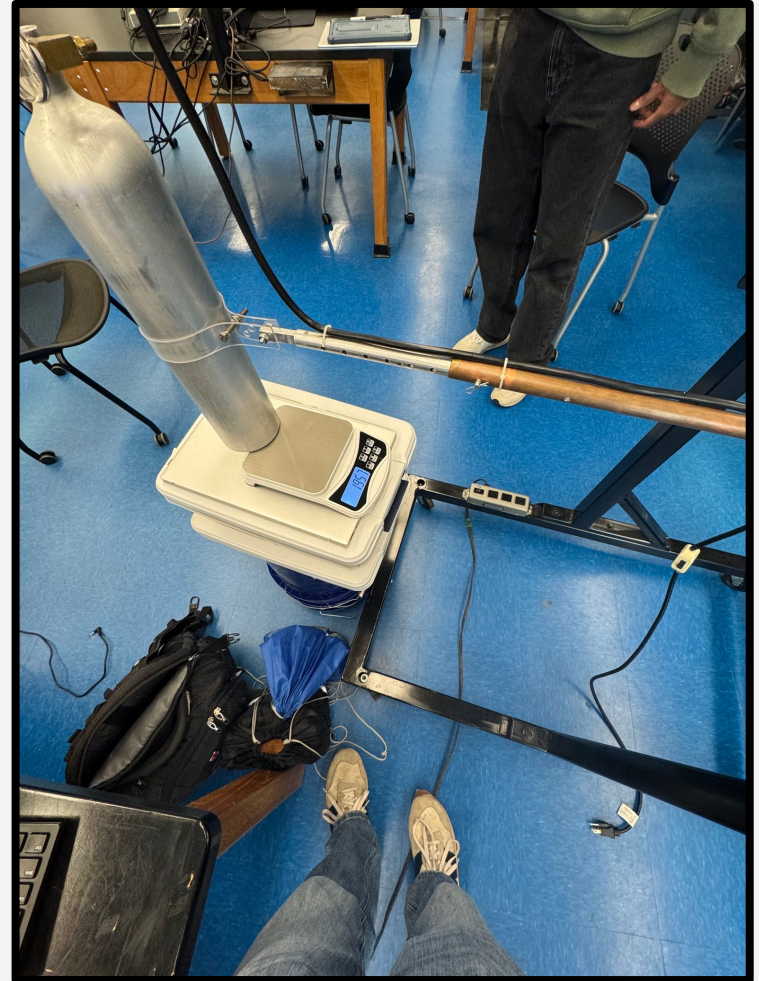
Procedure

Our testing apparatus attaches a pressurized air cylinder to a free pendulum, measuring the rotation through a potentiometer.

Initial mass of the apparatus was available online. Center of gravity distance was found using scale and moment balance

Free swing test was conducted to determine polar moment of inertia via period of oscillating system:

$$J = \frac{mgL_{CG}}{\left(\frac{2\pi}{T}\right)^2}$$



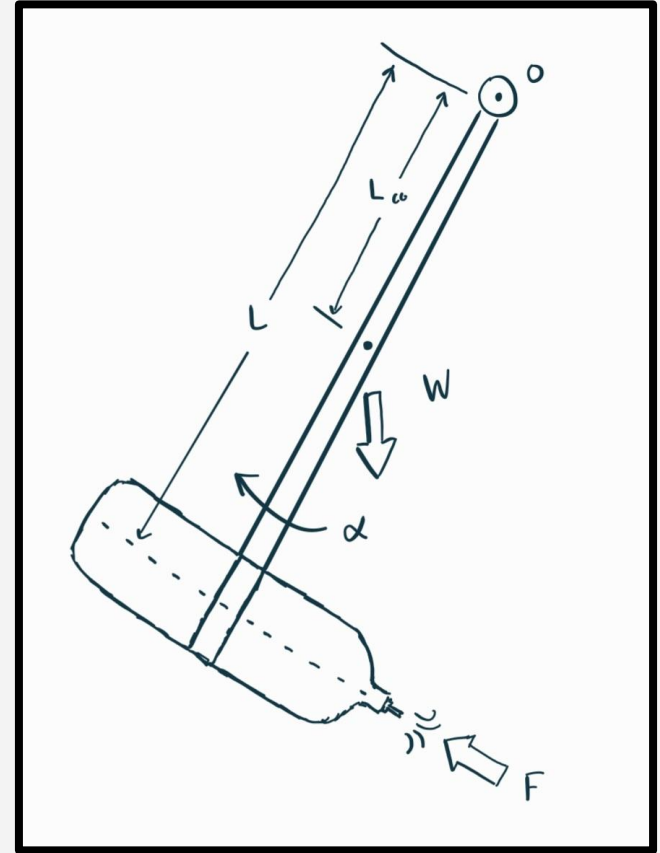
Procedure

The primary procedure is performed by using an air compressor to build pressure within the vessel to the desired level (measured with external gauges)

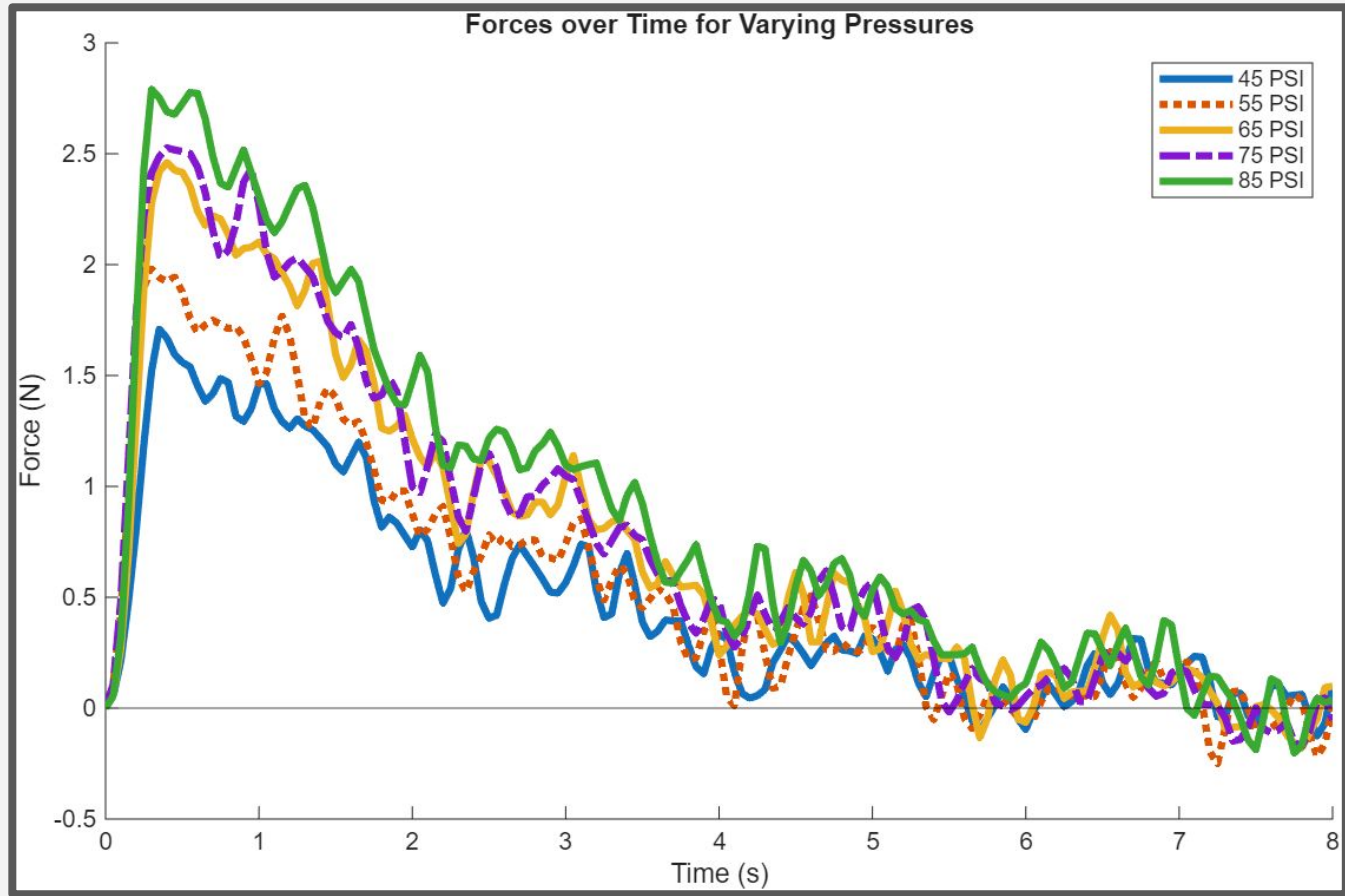
Afterwards, the cylinder is cleared of attachments and pressure is released via a solenoid from the equilibrium position, with its rotation captured as it moves.

The experimental force could be determined using a moment balance, based on angular acceleration and angle data from PASCO

$$F = \frac{J\alpha + mgL_{CG}\sin\theta}{L}$$

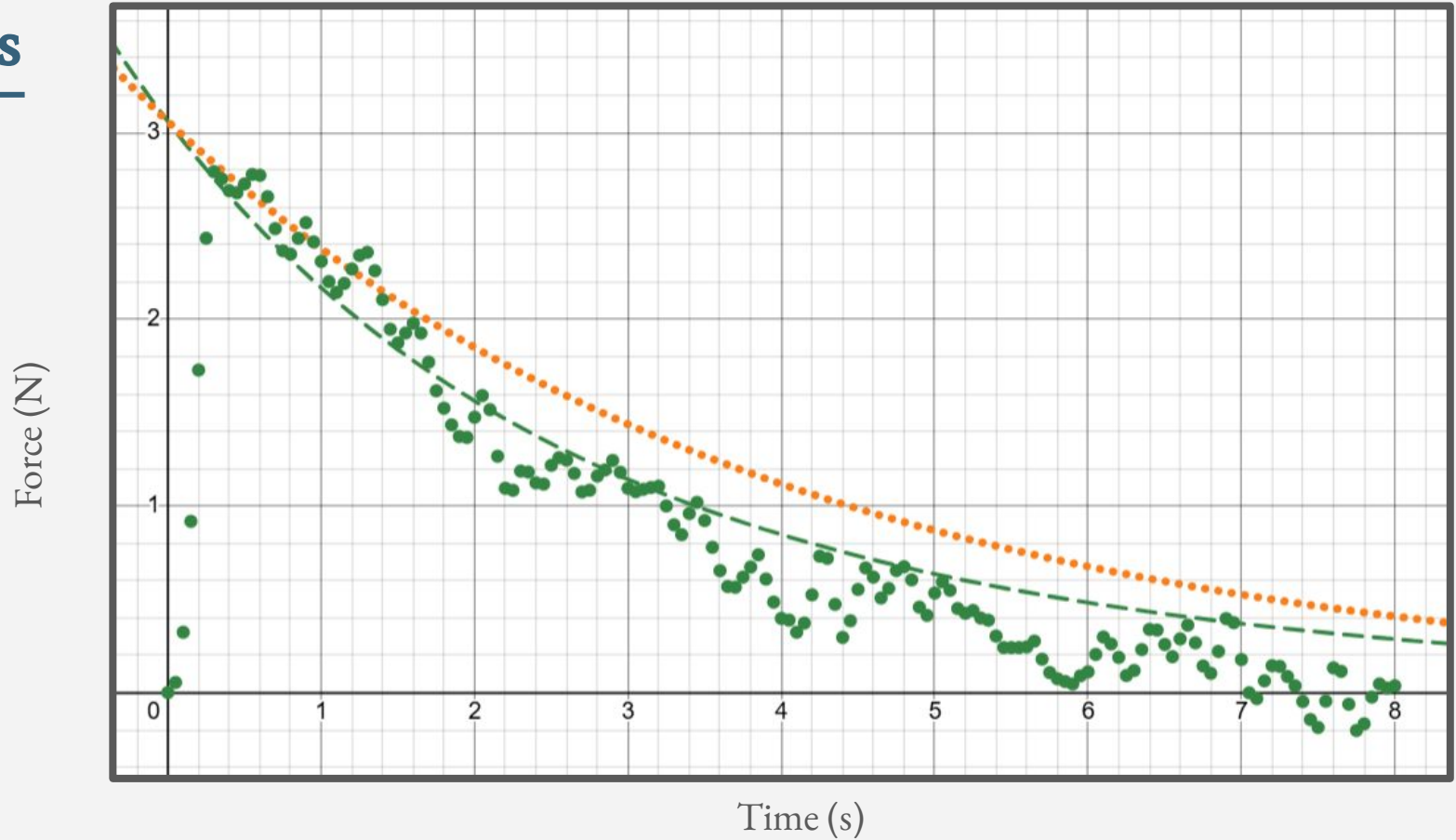


Results



Comparison of Adiabatic & Isothermal Models (85 psi)

Results

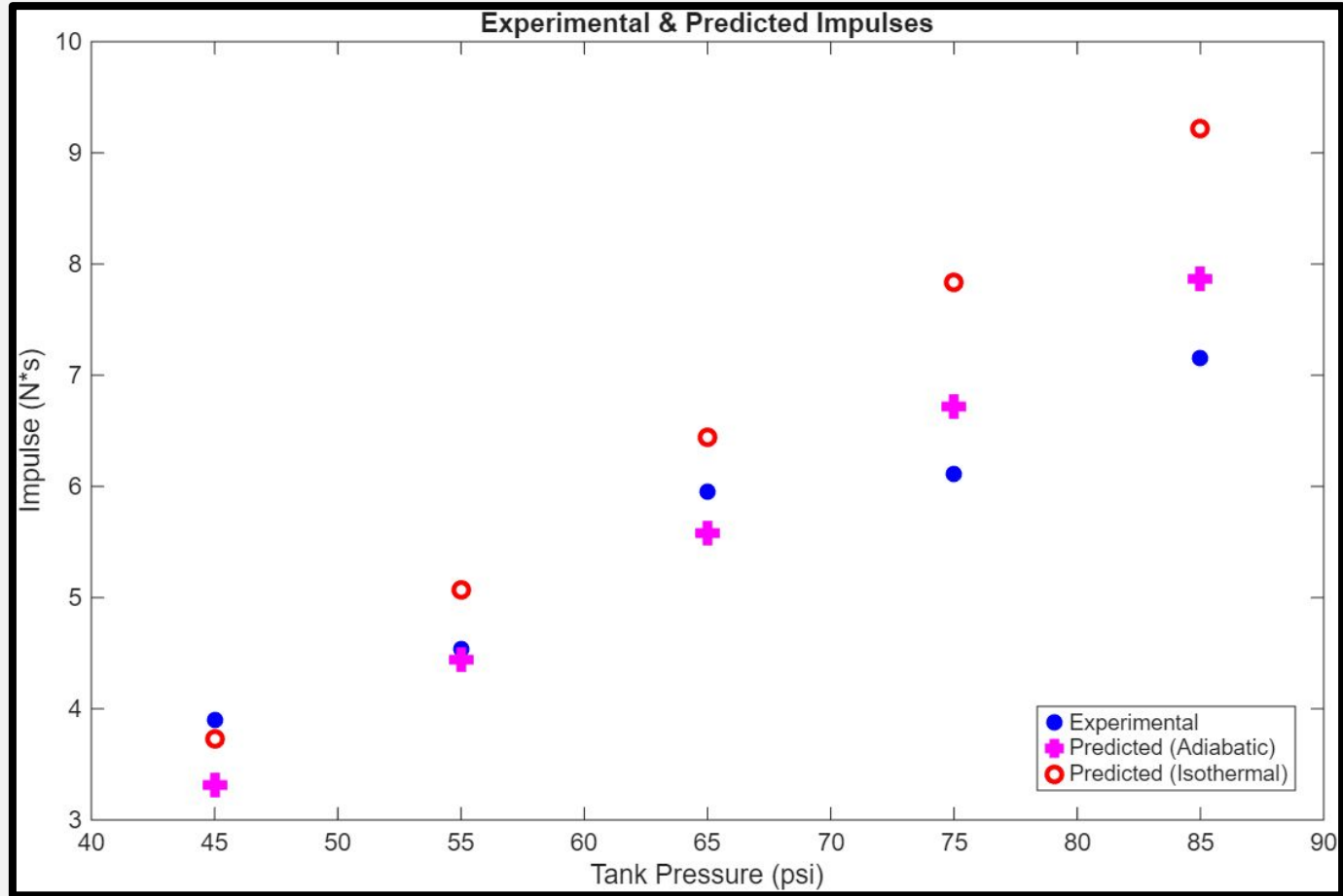


Results

Impulse Table for Varying Pressures

Tank Pressure (psi)	Impulse (N*s)		
	Experimental	Predicted (Adiabatic)	Predicted (Isothermal)
45	3.90	3.31	3.73
55	4.54	4.44	5.07
65	5.95	5.58	6.44
75	6.11	6.72	7.83
85	7.15	7.87	9.22

Results



Analysis

The results show a clear trend: as tank pressure increases, the impulse also increases; **higher pressure** means higher mass flow rate and exit velocity, so you get **more thrust**. This lines up with the original idea that most of the impulse happens right at the start when the pressure is highest

At higher pressures, the adiabatic model starts to overshoot but remains comparatively accurate ($\sim 10\%$).

The isothermal model is consistently higher than the experimental values, and the gap gets bigger as pressure increases. For example, 85 psi it predicts 9.22 N·s vs. a true value of 7.15 N·s. That tells us the system isn't behaving isothermally, since the air is **escaping too quickly** for heat transfer to keep the temperature constant.

The adiabatic is a **better fit**, especially in the middle pressure range but both models begin to drift at higher pressures.



Analysis

In all cases, adiabatic predictions correctly model the impulses to within

15%

Sources of Error:

- Neglecting unchoked flow release region
- Variations in volume or nozzle diameter measurement
- Inaccuracies in discharge coefficient selection
- Bearing friction at pivot point of system

Conclusion

The results show that impulse **increases steadily** with the tank pressure, confirming the validity of the test's expected relationship between pressure, mass flow rate, and thrust

The **adiabatic model** aligns with the experimental data across most pressures. On the other hand, the isothermal overpredicts impulse, showing that heat transfer is not sufficient to maintain constant temperature during the tank's release

In real life, the system is likely neither, requiring more **complex transient approach**

At higher pressures, both models start to **overestimate** impulse which could be due to **real world limitations**

References

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